Exploring Signed Null Models in Transcriptional Networks through Motif Analysis

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1 Introduction

Transcriptional regulation networks are a type of biological network that describes the interactions between regulators and their target genes. These networks are essential for understanding the regulation of gene expression and the control of cellular processes. In this project, we analyze the transcriptional regulation network of Escherichia coli (E. coli) K-12, a model organism widely used in microbiological research.

The network comprises three types of interactions: regulators and genes, regulators and transcriptional units, and regulators and other regulators. Regulators are proteins that control the expression of genes by binding to specific DNA sequences. Genes are DNA sequences that encode proteins, and transcriptional units are groups of genes transcribed together. The interactions between regulators and genes/transcriptional units can be positive (activator), negative (repressor), dual (both activator and repressor), or unknown. The interactions are also classified by their confidence level, which can be confirmed, strong, weak, or unknown.

The network study is done through the analysis of motifs, which are small subgraphs that appear more frequently in the network than expected by chance. Motifs are important because they can provide insight into the function of the network and the regulatory circuits that are present. In this project, we focus on 3-node motifs, which are connected subgraphs of three nodes. We analyze the motif occurrences in the network and compare them to different null models to assess their significance, highlighting the importance of choosing appropriate models for accurate interpretation and how this is linked to the signed nature of transcription networks.

2 Dataset description and preprocessing

All the data used for the project was obtained from RegulonDB (`http://regulondb.ccg.unam.mx/`). RegulonDB is a comprehensive database that provides detailed information about the regulatory networks of Escherichia coli (E. coli) K-12, a model organism widely used in microbiological research. The most updated version of the database is RegulonDB v12.0[1]. Among all the features contained in the database, the most relevant for this project are the transcriptional regulatory networks. Initially, the raw data is split in three files where each of them contains different types of interactions.

- **NetworkRegulatorGene**: This file contains the regulatory network interactions between regulators and their regulated genes
- **NetworkRegulatorTU**: This file contains the regulatory network interactions between regulators and their regulated transcriptional units
- **NetworkRegulatorRegulator**: This file contains the regulatory network interactions between regulators and other regulators

Each of the files can be downloaded as a `.csv` file and contains the following columns:

- **regulatorId**: The unique identifier of the regulator
- **regulatorName**: The name of the regulator

- **RegulatorGeneName**: The name of the gene that encodes the regulator
- **regulatedId**: The unique identifier of the regulated gene
- **regulatedName**: The name of the regulated gene
- **function**: Regulatory function of the regulator on the regulated gene (+ activator, - repressor, -+ dual, ? unknown)
- **confidenceLevel**: confidence level based on its evidence (Values: Confirmed[C], Strong[S], Weak[W], Unknown[?])

From each of the three files, we obtain a directed graph where the nodes are labelled with the unique identifier of the gene/regulator/transcriptional unit and also contain whether they are a gene, regulator or transcriptional unit. The edges are directed and contain two attributes: the confidence level of the interaction and their function.

The first step in the preprocessing of the data is to merge the three networks into a single one. This is done by first aggregating the nodes of the three networks and then adding the edges. Then, as a second preprocessing step we remove all the edges that have a level of confidence different from "Confirmed" or "Strong". This is done to ensure that the network is as reliable as possible, which will be important for the analysis of the motifs.

After the preprocessing, the network contains 4628 nodes and 10908 edges. It can be seen in Figure 1.

3 Basic network analysis

Before analyzing the motifs, it is important to understand the basic properties of the network. The first thing to notice is that the network is composed of 19 connected components, meaning that 19 regulatory circuits are independent of each other. We also counted the number of self-loops in the network, which are edges that connect a node to itself and there are 136 of them, which correspond to transcription factors that regulate themselves. The network density is 0.001, which means that only 0.1% of all possible edges are present in the network. This is a very low density and it is expected since the transcriptional regulation networks tend to be sparse.

The degree distribution of the network is shown in Figure 2. It can be seen its power-low behaviour, meaning that the transcriptional regulation network of E. coli K-12 is a scale-free network. This is a common property of biological networks and it means that there are a few nodes with a very high degree and many nodes with a low degree. The node with the highest degree is the regulator CRP (ID: RDBECOLITFC00189) which has an out-degree of 955 and an in-degree of 4. CRP (Catabolite Repressor Protein) is a global regulator that controls the expression of many genes in response to the presence of glucose. This protein forms part of the core-genome of many bacteria species (defined as the gene sequences present in all members of a bacterial species)[3].

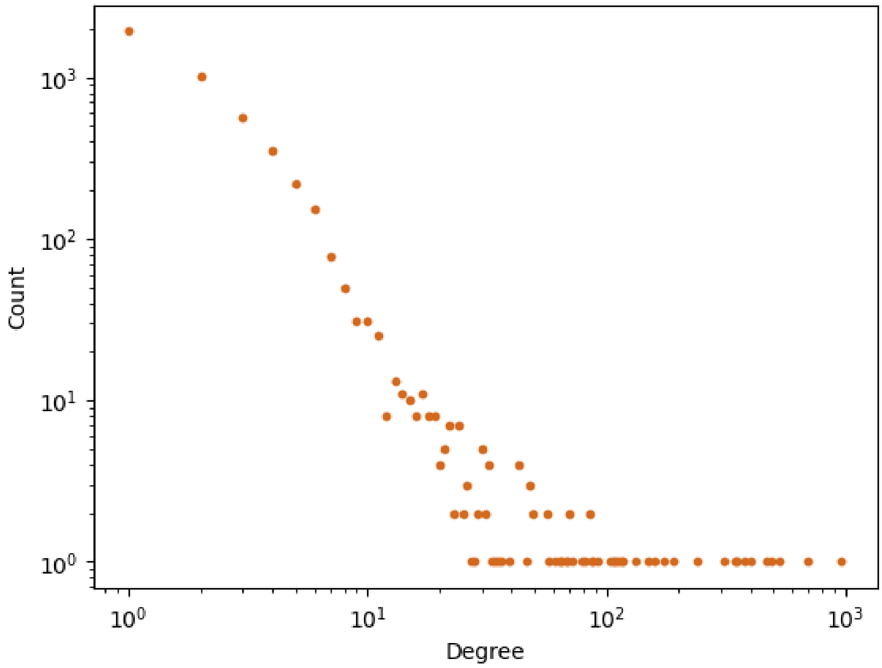

Figure 2: Degree distribution of the transcriptional regulation network of E. coli K-12.

Finally, we also compute the average clustering coefficient and the transitivity index of the network. The average clustering coefficient is 0.193 and the transitivity index is 0.008. The average clustering coefficient is a measure of the local connectivity of the network and the transitivity index is a measure of the global connectivity of the network. The difference between these two values is pointing out that the local structures of the network are much more connected than the global structure. This exposes the modular nature of the network, where many small groups of nodes are highly connected between them but not with the rest of the network, which emphasizes the importance of the motifs analysis.

4 Motifs analysis

In this section, we will analyze the 3-node motifs of the network. A motif is a subgraph that appears more frequently in a network than it would be expected by chance. They are important because they can give us information about the function of the network and in the case of transcriptional regulation networks, about the regulatory circuits that are present in the network.

In this section, we will disregard the sign of the interactions between the nodes and we will only consider the topology of the network. However, we will recover the signs of the interactions in the next section. There are 16 different types of triads that can be formed with 3 nodes. They are shown in Figure 3.

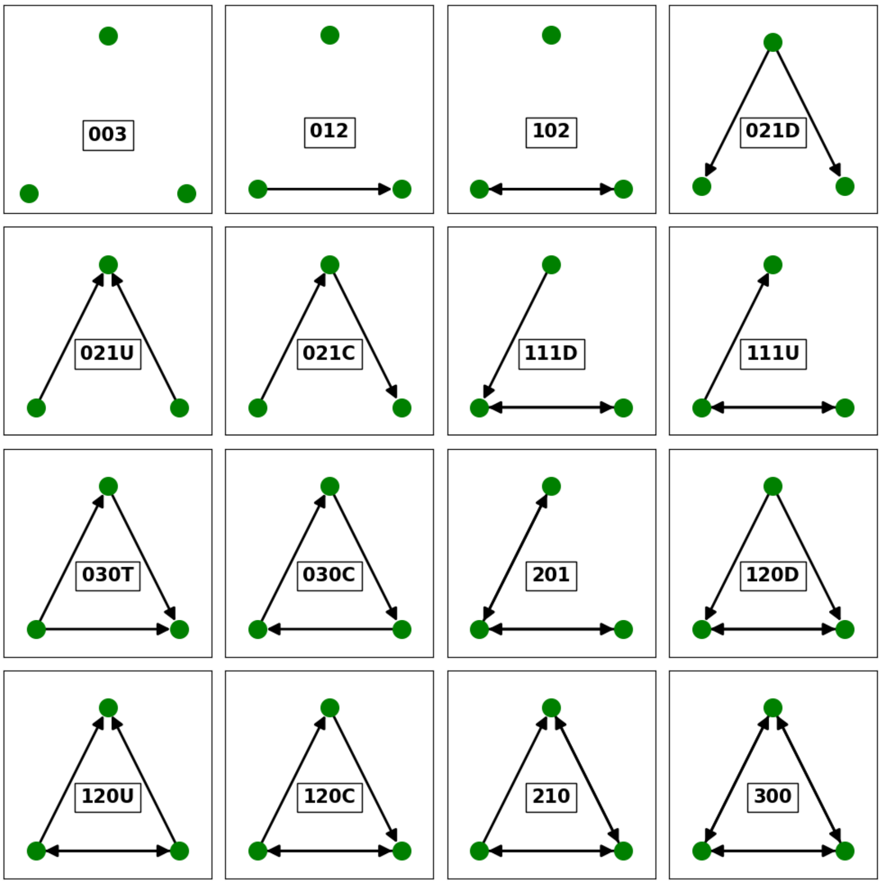

Figure 3: The 16 different types of 3-node motifs. Figure from NetworkX documentation[4].

From all the 3-node motifs, we will discard 003, 012 and 102 since they are not connected and they are not relevant for the analysis of the network. From the rest of motifs, we will distinguish between two subtypes, the 2-link motifs and 3-links motifs. The 2-link motifs are the ones that when converted to an undirected graph only have 2 edges (021D, 021U, 021C, 111D, 111U, 201). The 3-link motifs are the ones that when converted to an undirected graph have 3 edges (030T, 030C, 120D, 120U, 120C, 210, 300).

4.1 Efficient algorithm for motif counting

The next step in our analysis is to count the number of occurrences of each of the 3-node motifs in the network and for the next section we will also need to save a dictionary where the keys are the motif labels and the values will be all the triads of nodes that form the corresponding motif. One can imagine that the approach of running over all the possible 3-node subgraphs of the network and labelling them with one of the 16 motifs types shown in Figure 3 is not computationally possible. This is because the number of 3-node subgraph that one would need to check is given by the binomial

$$\binom{N}{3} = \frac{N!}{3!(N-3)!} = \frac{4628!}{3!(4625)!} = 16508750329 \approx 1.65 \times 10^{10} \tag{1}$$

Running over all these subgraphs would be computationally infeasible. However, once we notice the small density of the network we can think of a more efficient algorithm. The approach we design is the following:

For 2-link motifs:

- We run over the edge list of the network.
- For each edge we save the 2 nodes that form that edge.
- We compute and save in 2 lists all the neighbors of each of the nodes.
- We compute separately the intersection and the union of the neighborhoods.
- We save in a list all the nodes that belong to the union but not to the intersection.
- All the nodes in this list form a 2-link motif with the edge.
- Once we have all the triads we classify them in the corresponding motif.

For 3-link motifs:

- We run over the edge list of the network.
- For each edge we save the 2 nodes that form that edge.
- We compute and save in 2 lists all the neighbors of each of the nodes.
- We compute the intersection of the neighborhoods.
- All the nodes in the intersection form a 3-link motif with the edge.
- Once we have all the triads we classify them in the corresponding motif.

Notice that this approach allows us to classify the motifs in $O(E)$ time, where E is the number of edges in the network, which makes us move from the $O(10^{10})$ to a $O(10^4)$ complexity. This is a huge improvement and allows us to compute the motifs in a reasonable time.

4.2 Null models

Once we have the motifs classify we can simply count the number of occurrences of each of them in the network. However, this number is not informative by itself. We need to compare it with a null model. A null model is a random network that preserves some properties of the original network but that is otherwise random.

In the course bibliography[5] they use an Erdös-Rényi random network as a null model. This is a network where the edges are placed randomly between the nodes with a fixed probability. This network preserves the number of nodes and edges of the original network. Nevertheless, one of the most significant drawbacks of using Erdös-Renyi random network as a null model is the fact that its degree distribution is not representative of any real network. For this reason, in addition to the Erdös-Rényi random network, we will also use a configuration model as a null model.

The configuration model is a null model for networks that allow us to generate random graphs with a given degree distribution. In this model, the exact degree of each node is preserved and the edges are placed randomly between the nodes. However, there are different complexities of the configuration models that we can design. As a first approach, we will use a directed configurational model where we will preserve the in-degree and out-degree of each node. Finally, we will also use a signed directed configuration model that we will build from four degree sequences that need to be preserved: the in-degree of the positive nodes, the out-degree of the positive nodes, the in-degree of the negative nodes and the out-degree of the negative nodes.

We will compare the results of the real network with 100 realizations of each of the three previously described null models. We will compute the z-score of the number of occurrences of each motif in the real network with respect to the null models in order to make a comparison.

We show the results in Figure 4 and Figure 5. In Figure 4 we show the results of the motif count of the real network next to the directed (not signed) configurational model and in Figure 5 we show the results of the motif count of the real network next to the Erdös-Rényi random network and the directed configurational model.

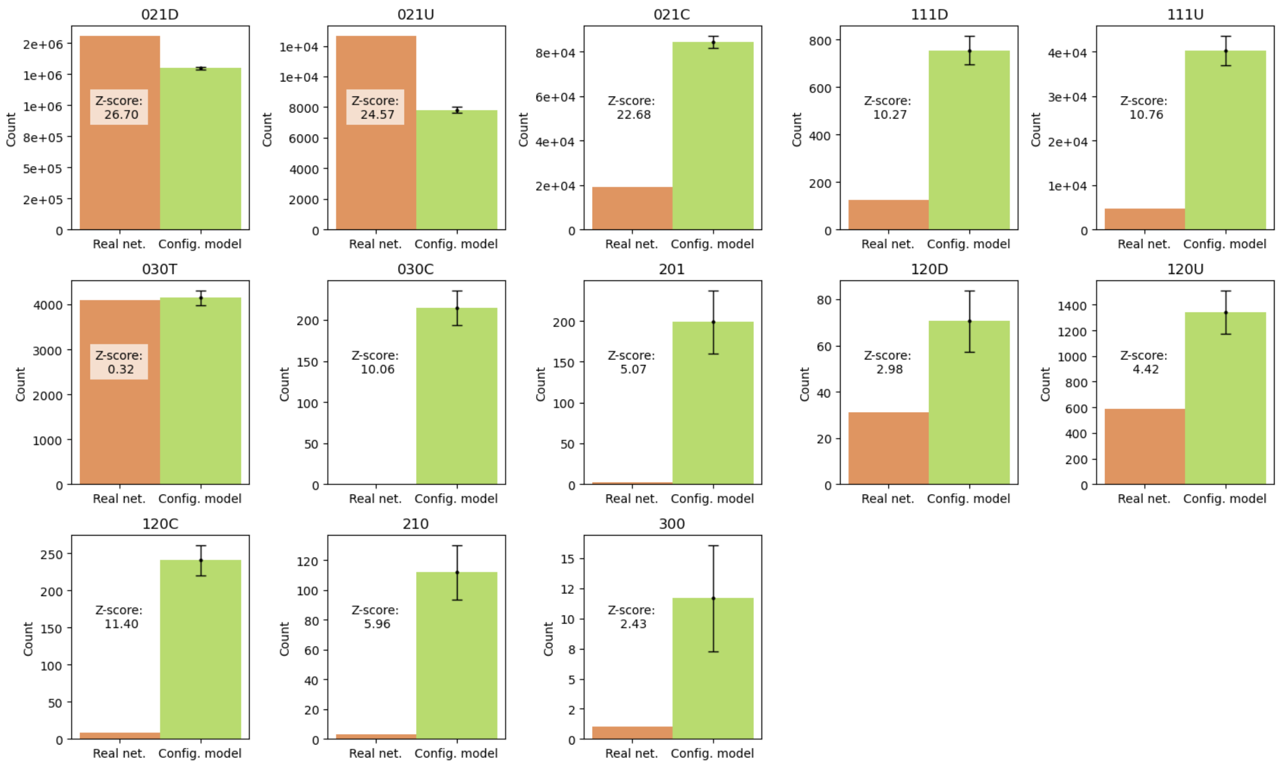

Figure 4: Motif count of the real network compared to the directed configurational model.

We can observe that the most abundant motifs in the real network are 120D, 120U and 120C. One might think that this is merely due to the simplicity of the nodes and the probability of forming a 3-link motif. However, when we compare the results with all the null models we can see that the number of occurrences of these motifs is significantly higher than what we would expect by chance, with the exception of motif 021U compared to the Erdös-Renyi, The most common motif, 021D, gives z-scores of 26.70, 15.39 and 6321.12 (directed configurational model, Erdös-Rényi random network and directed configurational model, respectively).

For the rest of the motifs we can observe that the conclusion drastically changes depending on the null model that we use since most of them have a larger significant occurrence in the real network compared to the Erdös-Rényi random network - which coincides with the results presented in [5] - but not compared to both of the configurational models. This emphasizes the importance of choosing the right null model for the analysis of the motifs since the conclusions that we can draw from the analysis can be very quantitative and qualitatively different.

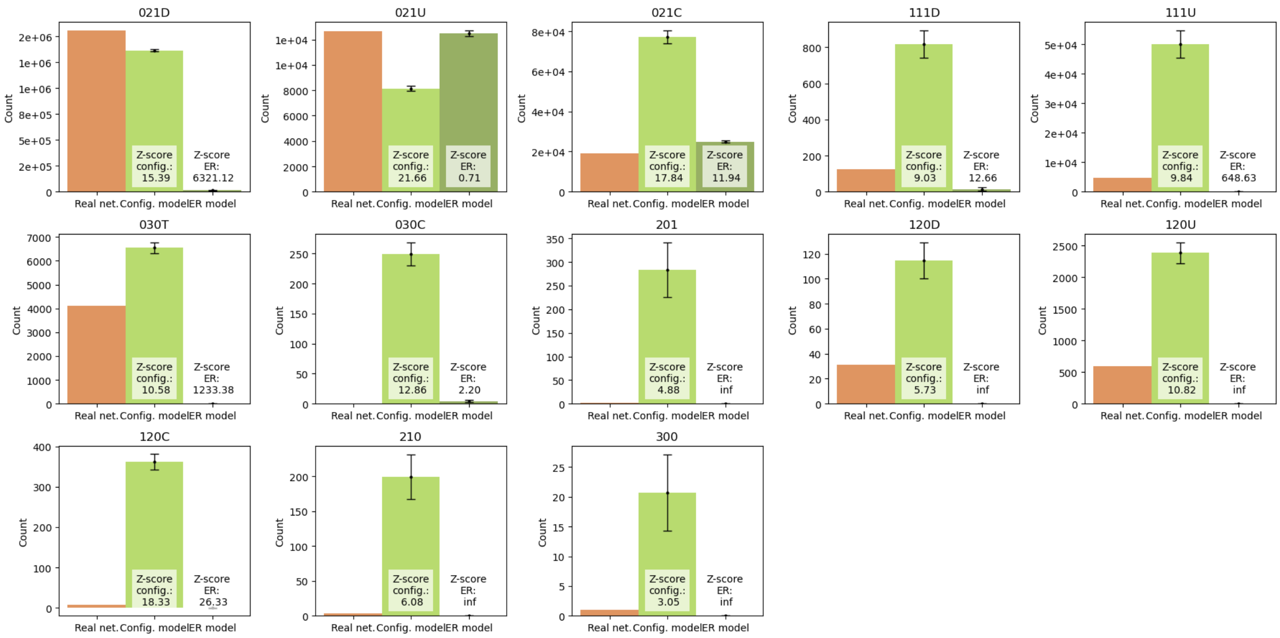

Figure 5: Motif count of the real network compared to the Erdös-Rényi random network and the signed directed configurational model.

The most interesting case is the motif 030T, which represent different types of feed-forward loops. In literature, feed-forward loops are known to be important for the robustness of the network and for the control of gene expression. In our case, we can see that the number of occurrences of this motif is significantly higher than what we would expect for the Erdös-Renyi random network, smaller than what we would expect for the signed directed configurational model and not significantly different from the directed configurational model.

The null model closer to the real network is the signed directed configurational model, which is expected since it preserves more properties of the network and the fact that the count of 030T motifs goes from "not significantly different from the null model" to "significantly below the null model" when we change from the directed configurational model to the signed directed configurational model is an example of the importance of the sign of the interactions in the motifs analysis. For this reason, we realize that it is essential to go a step further and analyze this motif accounting for the sign of the interactions.

5 Signed motifs analysis

In this section, we analyze the 3-node signed 030T motifs of the network. Since 030T has 3 edges and each of them can be positive or negative there are $2^3 = 8$ different types of 030T motifs. They are shown in Figure 7. A signed directed motif is considered balanced or coherent if, for every path that starts and ends at the same node, the product of the signs of the edges along that path is consistent.

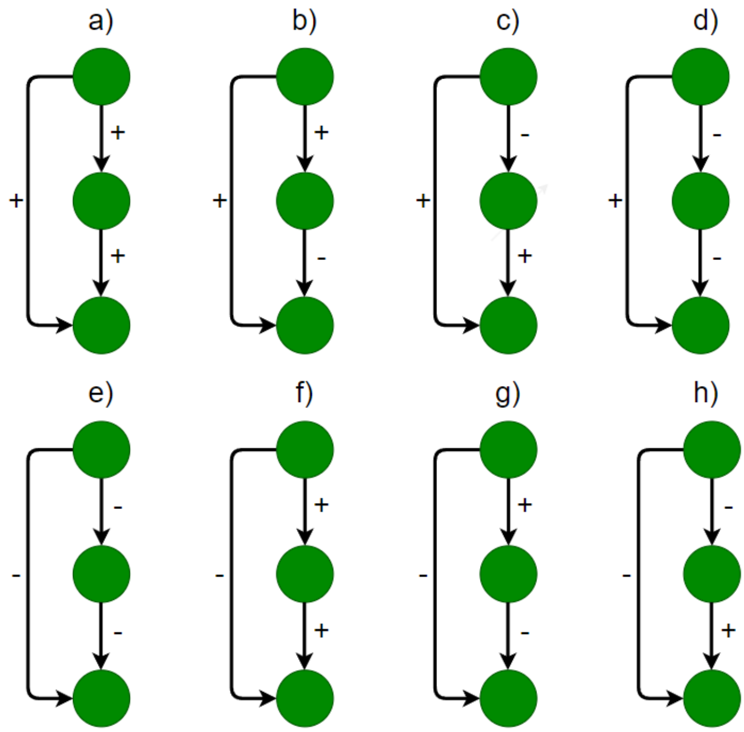

Figure 6: The 8 different types of 3-node signed 030T motifs.

If we consider nodes i and j connected through different paths P_1, P_2, ..., P_n, where each path is a collection of edges $e_1, e_2, ..., e_n$ with $e_1 = (i, k_1)$ and $e_n = (k_n, j)$, then if the product of the signs of the edges along the path is given by:

$$\prod_{e \in P_1} sign(e) = \prod_{e \in P_2} sign(e) = ... = \prod_{e \in P_n} sign(e) \tag{2}$$

Then the motif is considered balanced or coherent. Otherwise, it is considered unbalanced or incoherent.

Therefore, in Figure 7 we can see that the motifs a, d, g and h are coherent, while b, c, e and f are incoherent.

The importance of the sign of the interactions in the motifs analysis is that it can give us information about the function of the network. Indeed, each of the 8 different types of 030T motifs has a different function that is specified in Ref. [6]. For example, the a motif performs a sign-sensitive delay upon S_x steps. This means that the response to on steps is delayed, but the response to off steps of S_x is not delayed. Another example can be the b motif, which performs a sign-sensitive acceleration upon S_x steps, meaning that it can be used when a fast response is needed.

The algorithm to count the signed motifs is done by iterating over all the triads of nodes that we labelled as 030T motifs in the previous section. For each of the triads we check the sign of the interactions between the nodes and we classify them in one of the 8 different types of 030T motifs. We also do this for the directed signed null model, since it is the only one in which we can preserve the sign of the interactions, and compute the z-score for the occurrences of each of the 8 motifs. The results can be seen in Figure 7.

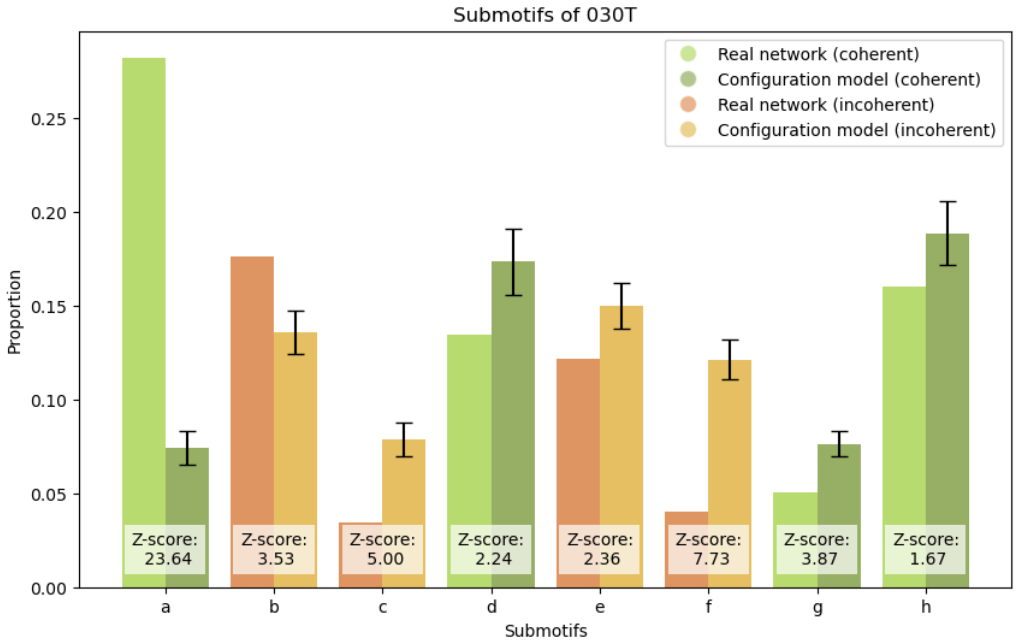

Figure 7

Except for type d the occurences of the rest of the motifs are significantly different from the null model. This is an interesting result that contrasts what we observed in Fig. 5, where the number of occurrences of the 030T motif was not significantly different from the null model. This is because there are significant differences between different types of 030T signed motifs occurences which are not captured by the total count of 030T motifs.

We can see that the most abundant motif in the real network is the a motif, and the second one is the b motif. Indeed these are the only two motifs whose occurrences are significantly larger than what we would expect by chance. This is in agreement with the results presented in [5], where they show that a and b motifs are the most abundant in the transcriptional regulation network of E. coli.

6 Conclusions

In this project, we analyzed the 3-node motifs of the transcriptional regulation network of E. coli K-12. Initially, we examined the motifs without considering the sign of the interactions, followed by an analysis of the signed motifs. We compared the real network's results with three different null models: the Erdös-Rényi random network, the directed configurational model, and the signed directed configurational model. Our findings revealed that the most abundant motifs in the real network are the 120D, 120U, and 120C motifs. Among the 3-link motifs, the 030T motif was the most prevalent.

When comparing our results to the null models, we found that the Erdös-Rényi random network is not a suitable null model as it can lead to qualitatively incorrect conclusions. The signed directed configurational model was the closest null model to the real network. When studying the occurrences of 030T signed motifs, we discovered that despite the total count of 030T motifs not being significantly different from this null model, the occurrences of the different types of 030T signed motifs are indeed significantly different, which higlights the importance of considering the sign of the interactions in the motifs analysis. The a and b motifs were the most abundant in the real network, and their functions are related to delay and accelerate the responses to signals, respectively.

In conclusion, analyzing the motifs of the transcriptional regulation network of E. coli K-12 is an important step towards understanding the network's function and regulatory circuits. The results of our analysis offer some conclusions about how we should interpret the motifs occurrences in the network. We show how the use of null models is essential in this analysis as they provide a comparative framework to assess the real network against a random network that preserves specific properties.

References

[1] Heladia Salgado et al. "RegulonDB v12.0: a comprehensive resource of transcriptional regulation in E. coli K-12". In: *Nucleic Acids Research* 52.D1 (Nov. 2023), pp. D255–D264. ISSN: 0305-1048. DOI: 10.1093/nar/gkad1072.

[2] Paul Shannon et al. "Cytoscape: a software environment for integrated models of biomolecular interaction networks". In: *Genome research* 13.11 (2003), pp. 2498–2504.

[3] Gloria Soberón-Chávez et al. "The Transcriptional Regulators of the CRP Family Regulate Different Essential Bacterial Functions and Can Be Inherited Vertically and Horizontally". In: *Frontiers in Microbiology* 8 (May 2017), p. 959. ISSN: 1664-302X. DOI: 10.3389/fmicb.2017.00959.

[4] Aric A. Hagberg, Daniel A. Schult, and Pieter J. Swart. "Exploring Network Structure, Dynamics, and Function using NetworkX". In: *Proceedings of the 7th Python in Science Conference*. Ed. by Gaël Varoquaux, Travis Vaught, and Jarrod Millman. Pasadena, CA USA, 2008, pp. 11–15.

[5] Uri Alon. *An Introduction to Systems Biology: Design Principles of Biological Circuits*. en. Google-Books-ID: tcxCkIxzCO4C. CRC Press, July 2006. ISBN: 978-1-58488-642-6.

[6] S. Mangan and U. Alon. "Structure and function of the feed-forward loop network motif". In: *Proceedings of the National Academy of Sciences of the United States of America* 100.21 (Oct. 2003), pp. 11980–11985. ISSN: 0027-8424. DOI: 10.1073/pnas.2133841100.

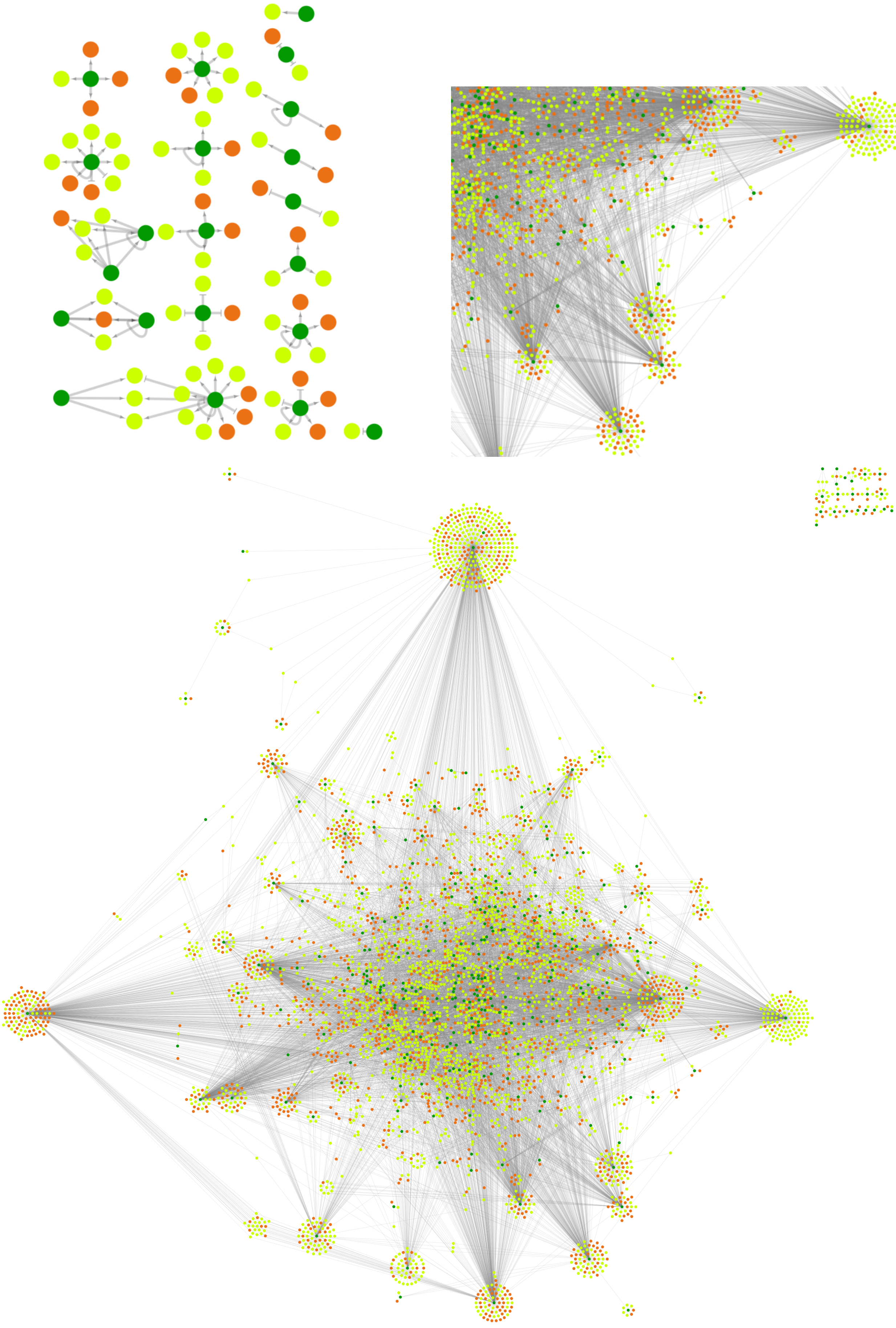

Figure 1: Network of the transcriptional regulation of E. coli K-12. The first two plots show a zoom of some regions of the network. The third plot shows the whole network. The plots were generated using Cytoscape[2].